LLaMA: Open and Efficient Foundation Language Models

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ABSTRACT

We introduce LLaMA, a collection of foundation language models ranging from 7B to 65B parameters. We train our models on trillions of tokens, and show that it is possible to train state-of-the-art models using publicly available datasets exclusively, without resorting to proprietary and inaccessible datasets. In particular, LLaMA-13B outperforms GPT-3 (175B) on most benchmarks, and LLaMA-65B is competitive with the best models, Chinchilla-70B and PaLM-540B. Unlike previous efforts which optimized for training FLOPs alone, our focus is on serving efficiency: a smaller model trained on more data is cheaper to run at inference time.

1. Introduction & The Inference Efficiency Imperative

Most recent work on large language models has focused on scaling parameter counts to hundreds of billions. However, recent scaling laws (Hoffmann et al., 2022) indicate that for a given compute budget, smaller models trained on significantly more data achieve superior downstream performance. Crucially, when considering total lifecycle cost, inference latency and hardware requirements dominate. We train models ranging from 7B to 65B parameters on 1.0T to 1.4T tokens of publicly accessible text.

2. Architectural Modernizations

LLaMA incorporates three key improvements over the standard Transformer:
1. Pre-normalization (RMSNorm): Normalize input of each transformer sub-layer using RMSNorm to improve training stability.
2. SwiGLU Activation Function: Replace ReLU non-linearity with SwiGLU activation (with dimension 2/3 * 4d) to boost model expressivity.
3. Rotary Positional Embeddings (RoPE): Employ rotary embeddings at each layer to improve relative position modeling and extrapolation.

3. Pre-training Data Pipeline & Tokenizer

The training corpus comprises 1.4 trillion tokens drawn exclusively from public sources: CommonCrawl (67%), C4 (15%), GitHub (4.5%), Wikipedia (4.5%), Gutenberg & Books3 (4.5%), ArXiv (2.5%), and Stack Exchange (2.0%). We utilize a Byte-Pair Encoding (BPE) tokenizer with 32,000 tokens, treating all numbers as individual digits to avoid arithmetic tokenization fragmentation.

4. Benchmark Performance Across Tasks

On common-sense reasoning benchmarks (BoolQ, PIQA, SIQA, HellaSwag, WinoGrande, ARC), LLaMA-13B surpasses GPT-3 (175B) despite being over 10x smaller. On MMLU, LLaMA-65B achieves 63.4%, trailing only Chinchilla-70B (67.5%) and PaLM 540B (69.3%). On GSM8K grade-school math, LLaMA-65B records 50.9%.

5. Limitations and Social Impact Disclosures

LLaMA models exhibit standard autoregressive vulnerabilities: propensity to generate hallucinations, potential toxicity generation without safety RLHF alignment, and linguistic over-representation of English (82% of pre-training tokens). Additionally, models were released as base weights without fine-tuned instruction following.

6. Future Directions

Open priorities include scaling token horizons beyond 2T tokens, fine-grained instruction alignment (Llama-2/3), and extending native context window lengths beyond 2048 tokens.